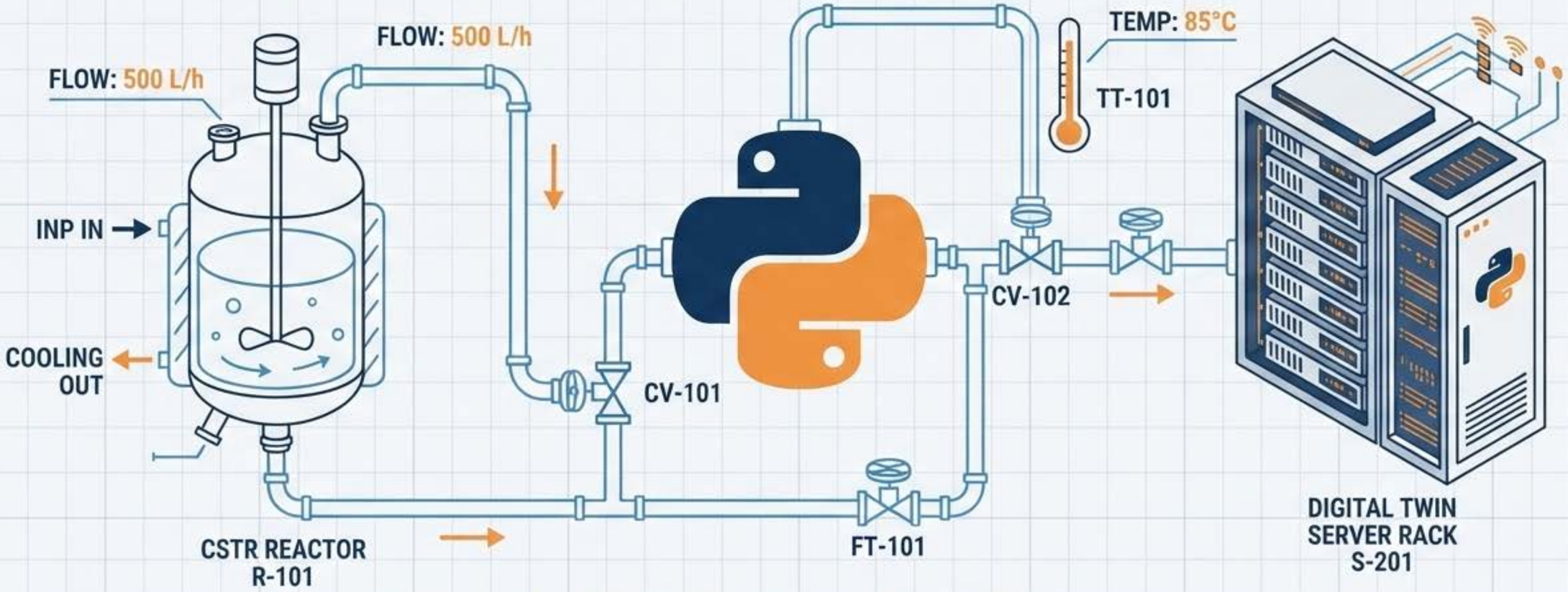
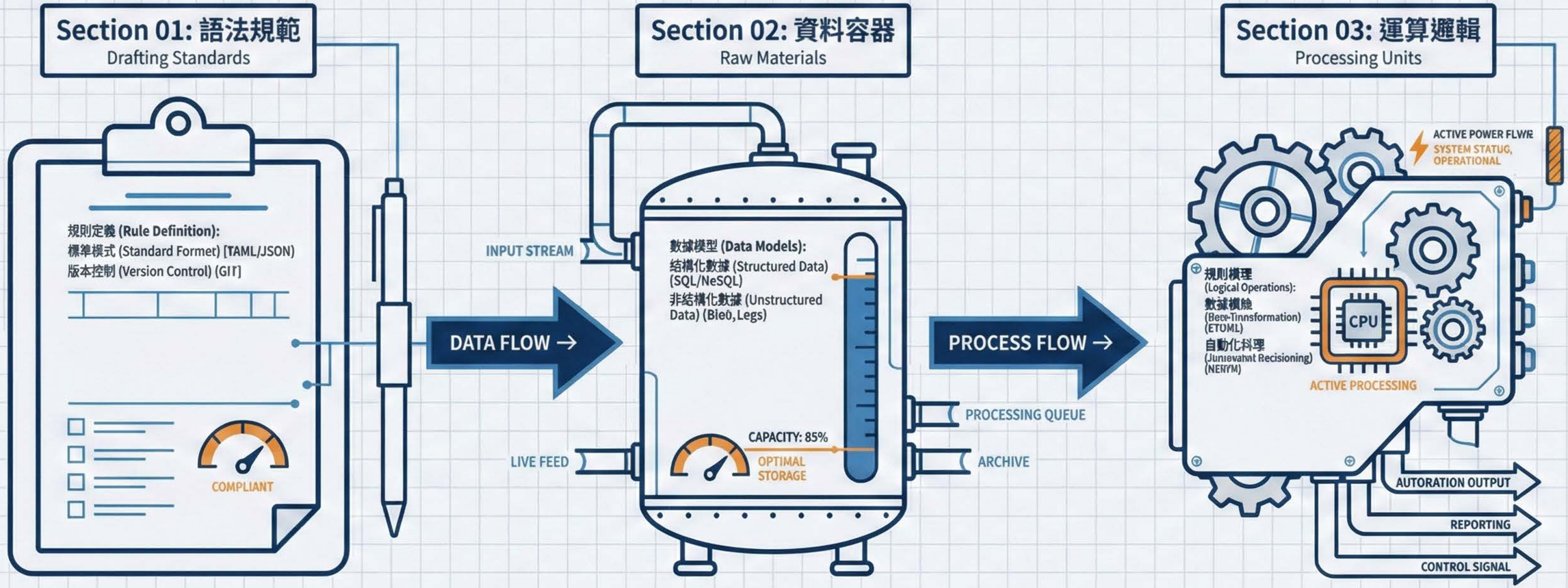
Unit 02 Python 程式語言基礎 (Part 1)

語法核心、變數結構與運算邏輯



建構數位化工廠的基石

數位化工廠的建構藍圖



NOTE

在自動化之前，我們必須先定義結構規則並了解材料特性。

程式碼的結構工藝：縮排 (Indentation)

Python 使用縮排
來定義區塊。
必須保持一致。

Good Code

```
Python = 6
for i = in():
    for i,z <=(0):
        # =tro["1]
        print("1")
        print("n")
return ()
```

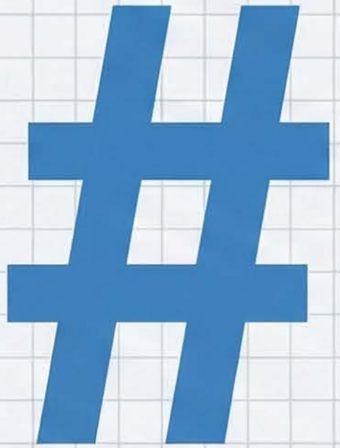
4 Spaces (4 個空格)

Bad Code

```
Python= 6
for i in=1():
   for i in <=(0):
      # =stro[""1]
       print("1")
    print("")
return ()
```

ALERT: IndentationError!
結構崩塌，無法執行

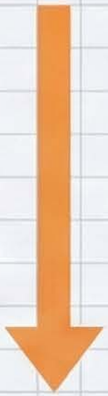
註解與執行順序：給人類看的操作手冊



註解(Comments)：程式碼中不會被執行的說明文字。

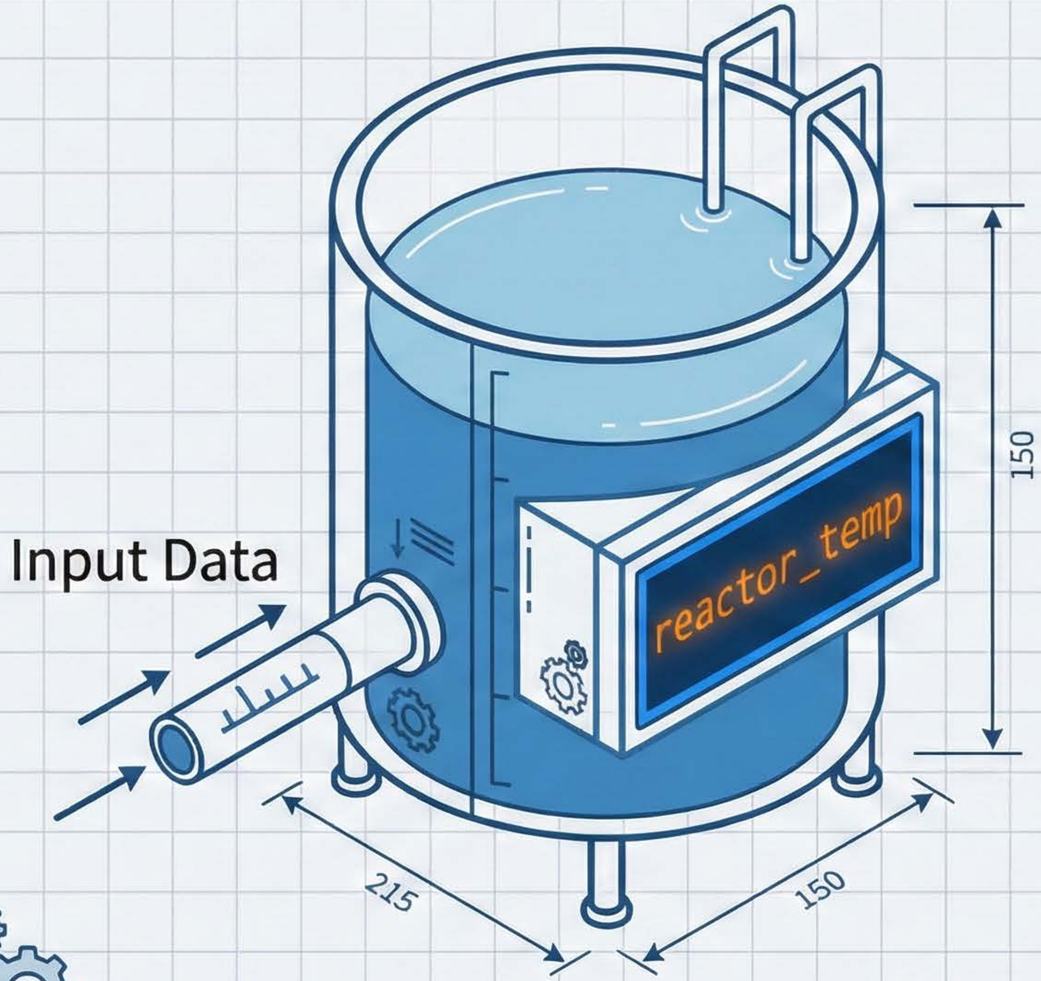
> Code is read much more often than it is written.
> (程式碼被閱讀的次數遠多於被撰寫的次數)

執行順序：由上而下 (Top-down)



```
# 計算反應器溫度
temp = sensor.read()
print(temp)
```

變數 (Variables)：數據的儲存容器

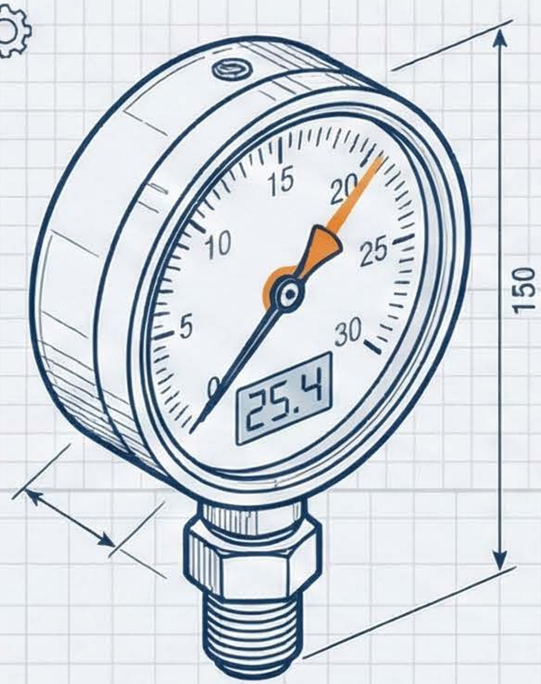


命名規則 (Naming Convention)

- **Snake Case:** 使用小寫與底線 (e.g., `reactor_temp`)
- **Descriptive:** 具有描述性 (避免使用 x, y)
- **Dynamic Typing:** 自動識別內容物 (無需宣告型別)

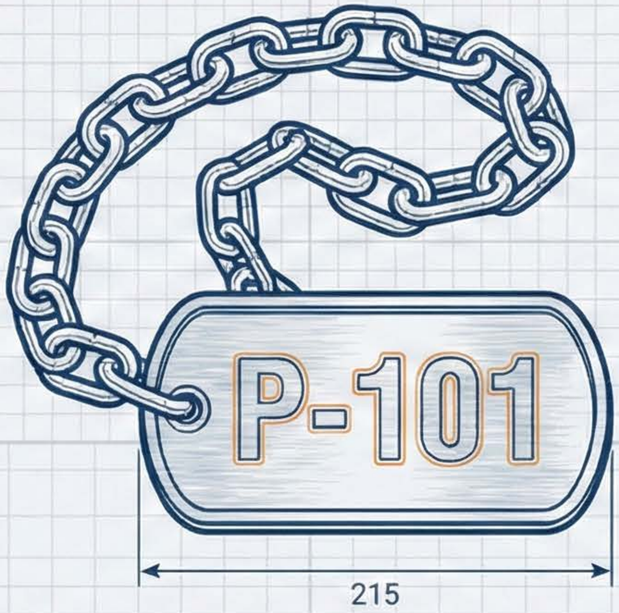
基礎原料：三大核心資料型態

數值 (Numeric)



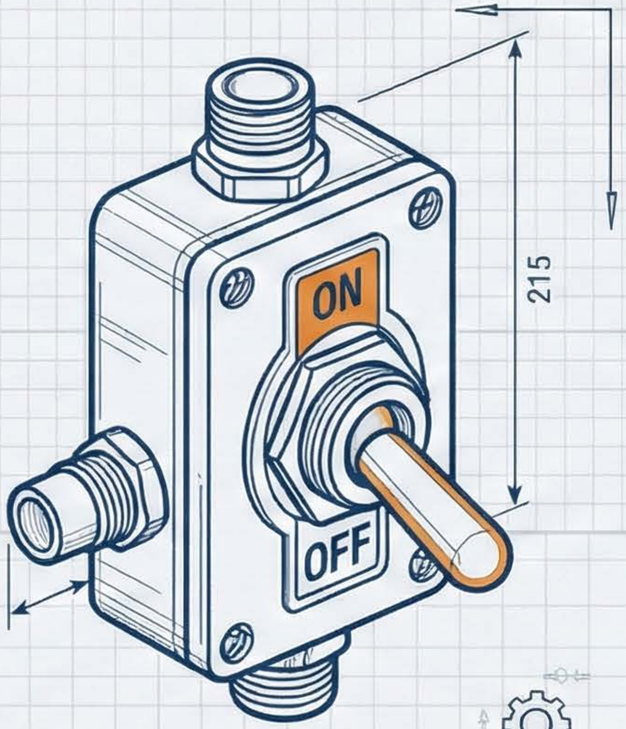
Float: 25.4
Int: 10

字串 (String)



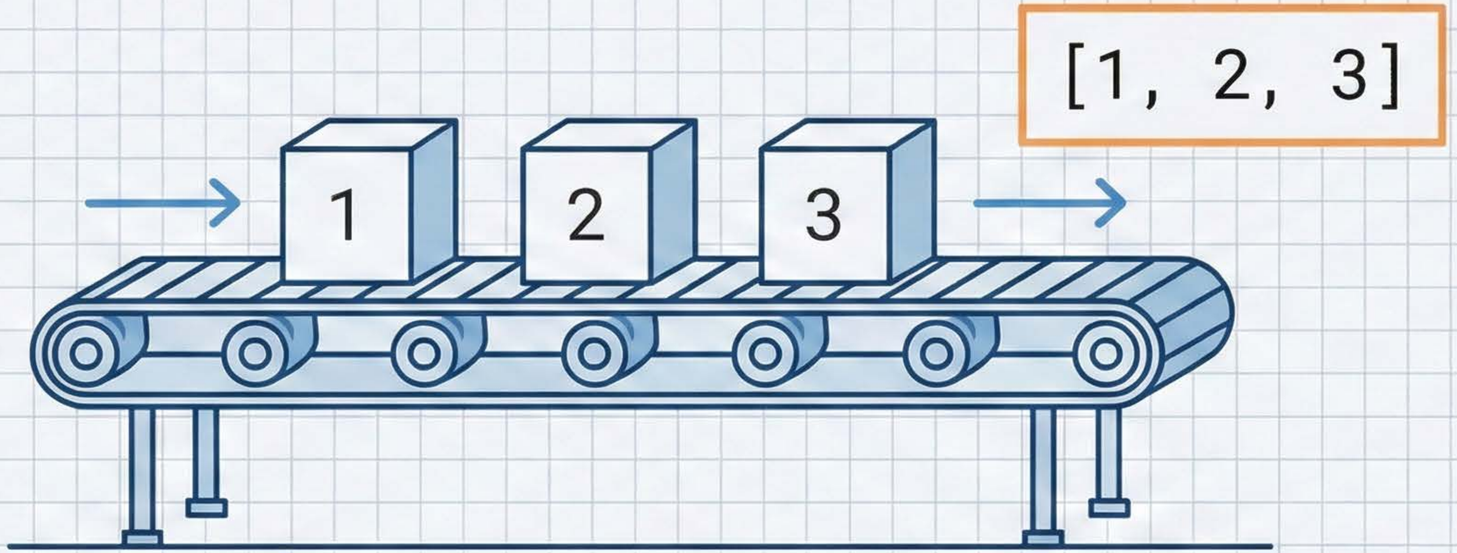
Tag_ID = ‘P-101’

布林值 (Boolean)



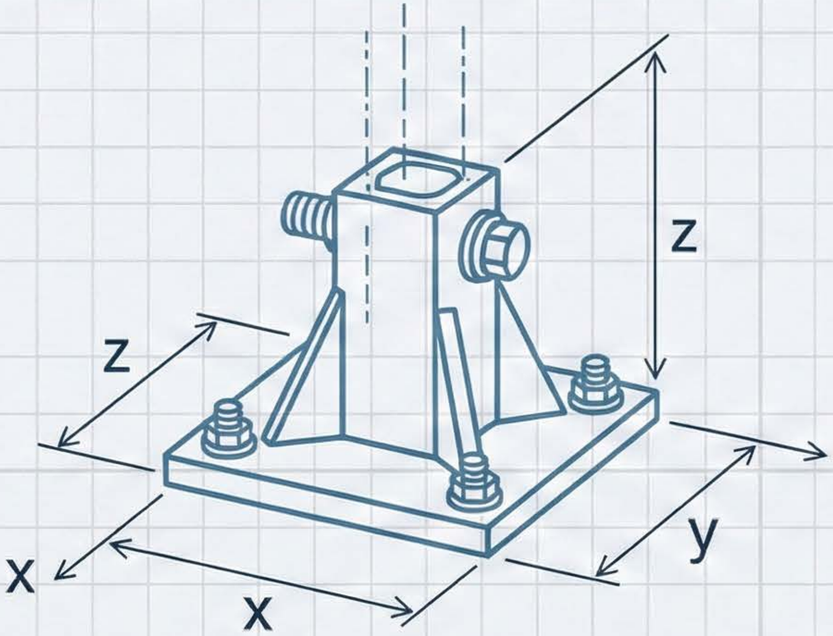
Valve_Open = True

串列 (List)：有序的可變輸送帶



- 符號：[ ] (中括號)
- Ordered (有序)：順序固定
- Mutable (可變)：內容可修改
- 應用：時間序列數據 (Time-series Data)

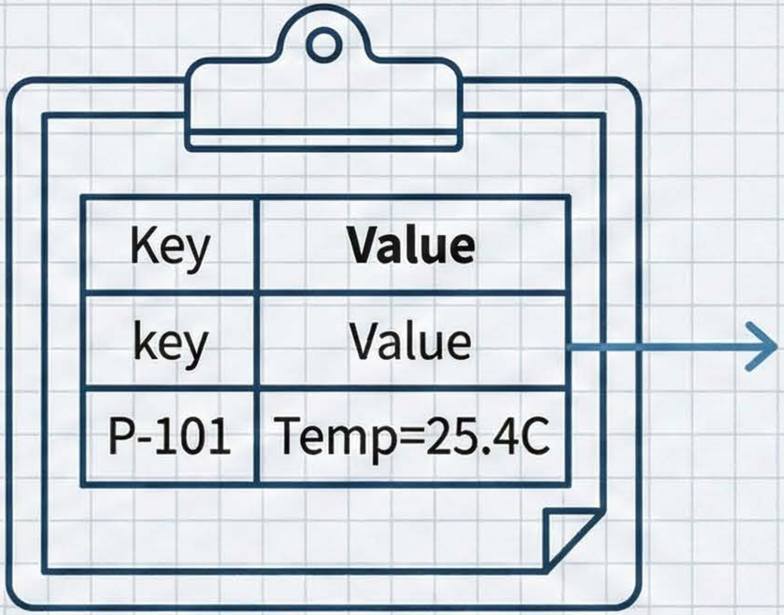
元組 (Tuple)：不可更動的設備參數



(x, y, z)

- 符號：( ) (小括號)
- Immutable (不可變)：定義後無法修改 (安全)
- Ordered (有序)：順序固定
- 應用：固定設備座標 (Fixed Coordinates)

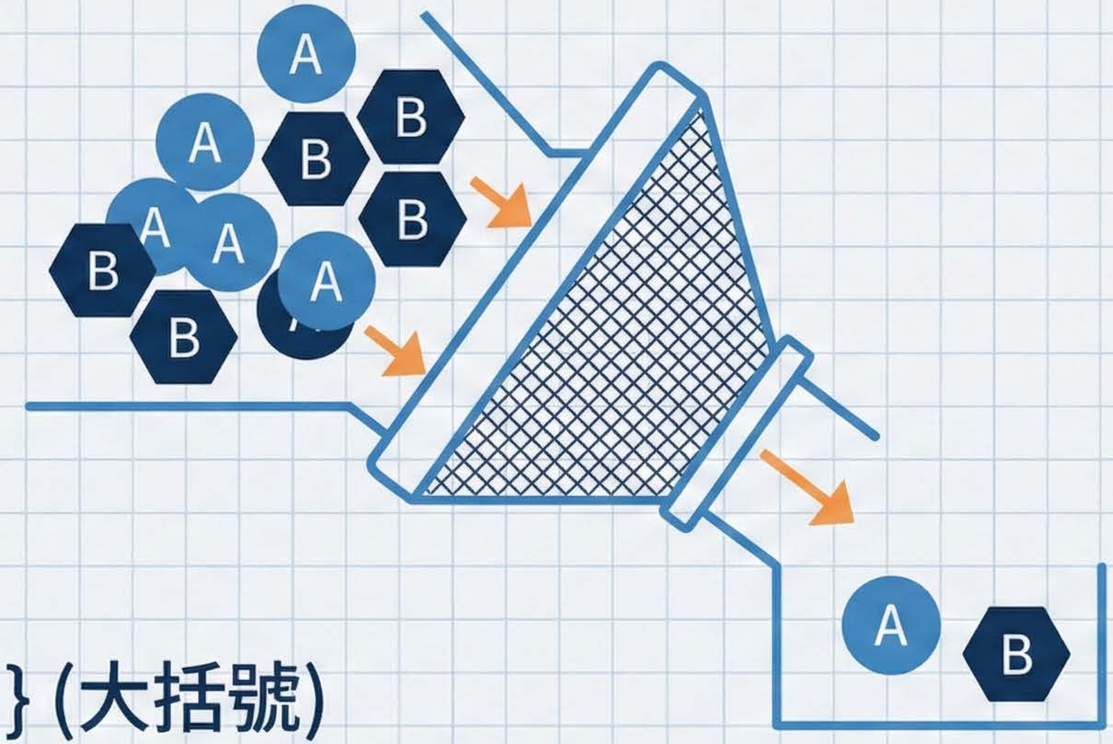
字典 (Dictionary)：鍵值對應的控制面板




```
{'key': 'value'}
```

- 符號：{k:v} (大括號+冒號)
- Key-Value (鍵值對應)：透過名稱查找
- Fast Lookup (快速查找)：類似物性查表
- 應用：設備屬性 (Equipment Properties)

集合 (Set)：過濾重複的篩選器



{'A', 'B'}

- ☑ 符號：{ } (大括號)
- ☑ Unordered (無序)：無固定順序
- ☑ Unique (不重複)：自動移除重複項
- ☑ 應用：錯誤代碼過濾 (Error Log Filtering)

資料結構選型指南 (Selection Guide)

結構 (Structure)	有序? (Ordered)	可變? (Mutable)	工業類比 (Analogy)
List (串列)	Yes	Yes	輸送帶 (Conveyor)
Tuple (元組)	Yes	No	固定設備 (Bolted Flange)
Dict (字典)	No	Yes	控制面板 (Control Panel)
Set (集合)	No	Yes	過濾器 (Filter)

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算術運算子：數值加工單元

+ 加法 Addition

− 減法 Subtraction

* 乘法 Multiplication

/ 除法 Division

**

Power (冪次方)

2**3 = 8

注意：不是 ^

%

Modulo (餘數)

10 % 3 = 1

(Remainder)

Calculation (計算)

Enthalpy = U + P * V

+=

Tank_Level += 10

(累積加總)

10

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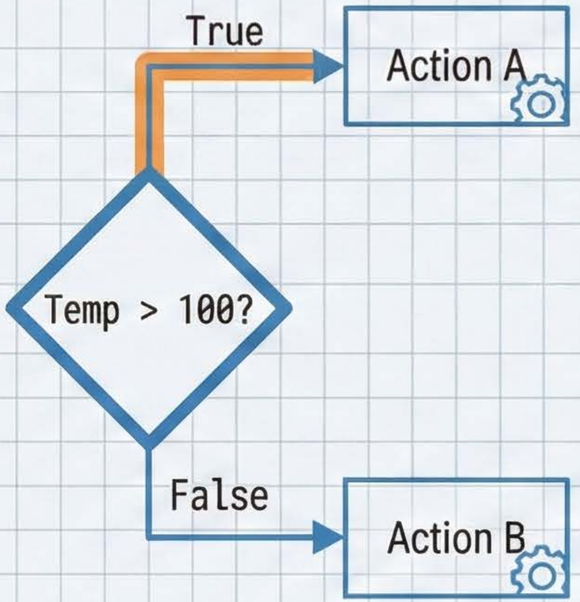
比較與邏輯：決策判斷的基礎

比較 (Comparison)

Operators:

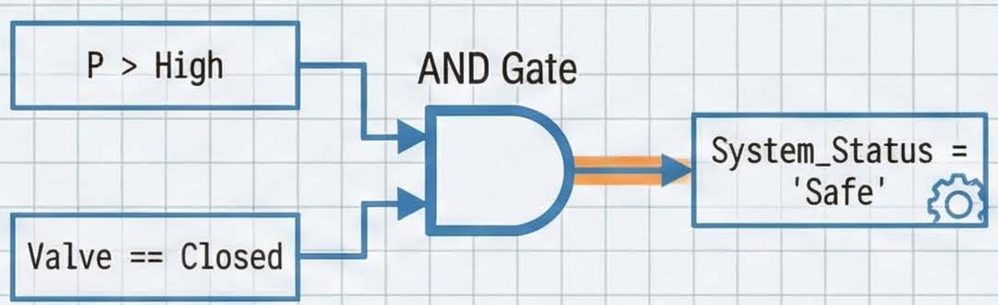
>, <, ==, !=;



邏輯 (Logic)

Operators:

and, or, not;



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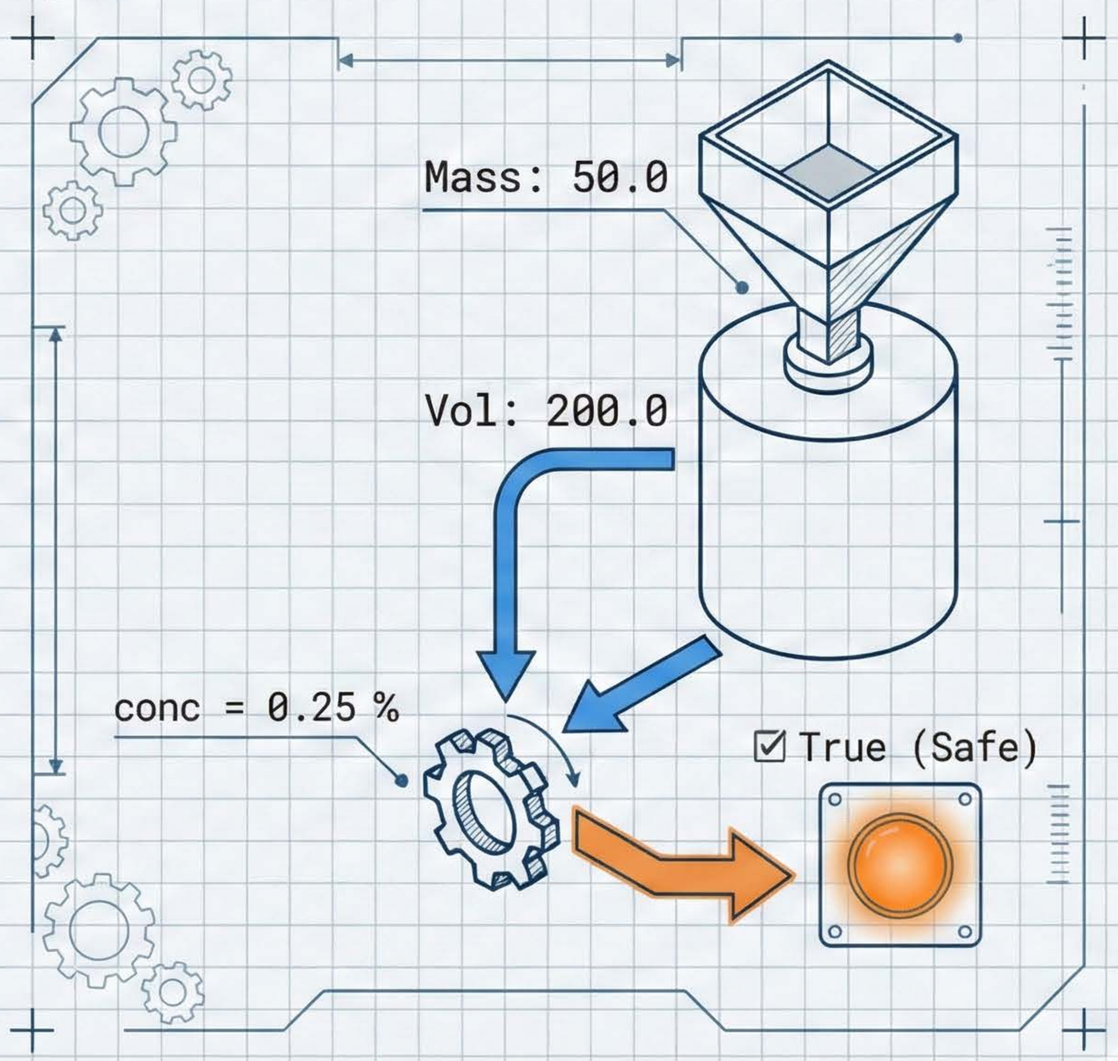
綜合應用：反應器參數計算

```
# 1. 定義變數 (Variables)
mass = 50.0
vol = 200.0

# 2. 算術運算 (Arithmetic)
conc = mass / vol

# 3. 邏輯檢查 (Check)
is_safe = conc < 0.5
```



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基礎已備，準備啟動自動化

Next Unit: Control Flow (流程控制)

下一步：將靜態藍圖轉化為自動化製程 (Conditions & Loops)

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